A Lightweight, Offline-Capable Bioinformatics Pipeline for kelch13 (K13) Artemisinin Resistance Surveillance in West African *Plasmodium falciparum*: Dual Validation on Simulated and Clinical Sequences

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Abstract

Background: Artemisinin-based combination therapies (ACTs) are the cornerstone of *Plasmodium falciparum* malaria treatment in West Africa [1, 2]. However, the global spread of artemisinin resistance, molecularly marked by non-synonymous mutations in the *P. falciparum* kelch13 (K13) propeller domain, represents a major public health threat [3]. Deploying lightweight, offline-capable bioinformatics surveillance tools in resource-constrained West African national reference laboratories is key to sustaining regional ACT therapeutic efficacy.

Methods: We developed a Python-based open-source pipeline that retrieves the 3D7 K13 reference sequence, constructs a local BLAST database, performs alignment-aware sequence parsing, and screens for the nine World Health Organization (WHO)-validated resistance mutations. To evaluate diagnostic performance, a dual-validation benchmark was established. First, an internal validation was performed using 30 *in silico* simulated West African sequences containing random validated point mutations. Second, an external validation was conducted on a geographically enriched clinical panel of 66 real West African K13 sequences (representing Benin, Ghana, Niger, Nigeria, and Senegal) retrieved programmatically from NCBI GenBank [4, 5, 6, 7]. To robustly process truncated Sanger amplicons and handle reverse-strand sequencing inputs without manual intervention, a 6-frame translation algorithm powered by a 5-mer match-scoring peptide heuristic was implemented.

Results: On the synthetic panel, the pipeline correctly classified all 30 isolates, yielding a mutation calling sensitivity and specificity of 100% (30/30 isolates), with zero false positives or false negatives. On the clinical panel, the 6-frame heuristic algorithm successfully translated and aligned all 66 sequences, resolving previous frame-shift and reverse-strand alignment anomalies (N/A percentages) with BLASTP identities ranging from 91.8% to 100.0%. The pipeline successfully called 65 clinical isolates as wild-type and automatically detected the presence of the WHO-validated Y493H resistance marker in a clinical isolate from Northern Ghana (accession OQ102665), replicating published clinical findings with high concordance [5].

Conclusions: This study validates a lightweight, highly resilient, and fully open-source pipeline capable of processing clinical Sanger sequences under realistic field conditions. Decoupling sequence translation and parsing from rigid absolute coordinates enables offline surveillance of emerging artemisinin resistance in Bénin and the broader West African region.

Keywords: *Plasmodium falciparum*; kelch13; artemisinin resistance; molecular surveillance; BLAST; bioinformatics pipeline; West Africa; Bénin.

1. Introduction

Malaria remains a leading cause of morbidity and mortality in sub-Saharan Africa, with *Plasmodium falciparum* accounting for the overwhelming majority of severe clinical cases and deaths on the continent. For over two decades, Artemisinin-based Combination Therapies (ACTs) have served as the undisputed first-line treatment for uncomplicated falciparum malaria across West Africa. The therapeutic efficacy of these combinations is, however, highly dependent on the rapid parasitological clearance driven by the short-acting artemisinin component.

Reduced clinical susceptibility to artemisinin derivatives, characterized by delayed parasite clearance, is molecularly associated with non-synonymous single nucleotide polymorphisms (SNPs) in the propeller domain of the *P. falciparum* kelch13 (K13, gene ID: PF3D7_1343700) [1]. A specific panel of these amino acid substitutions has been formally validated by the World Health Organization (WHO) as molecular markers of confirmed or candidate artemisinin partial resistance [1, 2]. While these mutations emerged and clonalized primarily within the Greater Mekong Subregion of Southeast Asia, the independent *de novo* emergence and subsequent clonal expansion of the K13 R561H mutation in Rwanda and DR Congo [3], alongside emerging candidate markers in the Horn of Africa, indicate that African parasite populations are actively selecting resistance alleles under drug pressure.

Molecular surveillance of K13 in West Africa is routinely carried out by national reference laboratories, often utilizing PCR amplification and Sanger sequencing of the propeller domain. However, translating these raw sequencer outputs into structured resistance reports presents significant logistical bottlenecks. Small laboratories often lack stable internet access, relying on remote international consortia [4] or cumbersome, expensive proprietary software to perform manually curated alignments. Furthermore, sequence deposits in public databases such as NCBI GenBank frequently consist of partial Sanger sequences, some of which are deposited as reverse-complement sequences relative to the coding strand [5, 6]. Naive translation of these reverse-strand reads or truncated sequences generates reading-frame shifts and premature stop codons, leading to alignment failures in standard automated pipelines.

To address these challenges, we designed, implemented, and validated an open-source, lightweight, and offline-deployable Python pipeline. This pipeline uses a resilient 6-frame translation heuristic and alignment-aware coordinate walking to screen West African K13 isolates for resistance mutations. We report a comprehensive dual-validation of the pipeline using both simulated datasets (internal validation) and a geographically diverse clinical panel of 66 real West African sequences (external validation) retrieved from GenBank [4, 5, 6, 7].

2. Materials and Methods

2.1 Reference Sequence Retrieval

The full-length reference protein sequence of *P. falciparum* 3D7 K13 (726 amino acids; NCBI RefSeq accession XP_001350158.1) was retrieved programmatically using the BioPython `Bio.Entrez` module. A local protein BLAST database was constructed from this single reference sequence using the `makeblastdb` command from the NCBI BLAST+ suite (v2.12.0) with the `-dbtype prot` configuration.

2.2 Resistance Mutation Panel

Nine WHO-validated K13 propeller domain mutations were compiled into a reference lookup table (Table 1), containing the 1-based reference position, wild-type residue, mutant residue, and primary geographic context. All nine positions fall within the Kelch propeller domain (approximate residues 440–680).

Table 1: WHO-validated kelch13 propeller domain resistance mutations.

Mutation	**Position**	**Wild-type**	**Mutant**	**Reported Primary Context**
F446I	446	Phe (F)	Ile (I)	Myanmar / Southeast Asia [1]
N458Y	458	Asn (N)	Tyr (Y)	Southeast Asia [2]
M476I	476	Met (M)	Ile (I)	Southeast Asia [1]
Y493H	493	Tyr (Y)	His (H)	Southeast Asia [1, 5]
R539T	539	Arg (R)	Thr (T)	Southeast Asia / emerging in Africa [2]
I543T	543	Ile (I)	Thr (T)	Southeast Asia [2]
P553L	553	Pro (P)	Leu (L)	Southeast Asia [1]
R561H	561	Arg (R)	His (H)	Rwanda / DR Congo (emergent in Africa) [3]
C580Y	580	Cys (C)	Tyr (Y)	Global / Dominant marker [1, 2]

2.3 Simulated (Internal) Validation Dataset

To evaluate the logical sensitivity and specificity of the mutation calling module, a synthetic panel of 30 mock isolates was constructed *in silico*. Sequences were generated by assigning random simulated West African countries of origin (Bénin, Burkina Faso, Côte d'Ivoire, Ghana, Mali, Nigeria, Senegal) to each record. Point mutations from Table 1 were introduced into 23.3% of the sequences (7/30), while the remaining 76.7% (23/30) retained the unmodified reference sequence. The ground-truth mutations were stored separately as a tab-separated file and hidden from the pipeline during execution.

2.4 Multi-Country Real (External) Clinical Dataset

To evaluate the clinical specificity of the pipeline under realistic surveillance conditions, we constructed an enriched geographical panel of 66 real clinical sequences from West Africa downloaded programmatically from NCBI GenBank. The panel was composed of Benin ($n = 20$; accessions MN072940 to MN072959), Niger ($n = 15$; accessions MZ364160 to MZ364174), Nigeria ($n = 15$; accessions MT113570 to MT113584), Ghana ($n = 15$; accessions OQ102653 to OQ102667), and Senegal ($n = 1$; accession MH464876).

To ensure that the clinical panel contained only genuine West African sequences, the script parsed the GenBank metadata programmatically. First, the source country was extracted from the `/country` qualifier of the GenBank record's source feature. Second, if this qualifier was absent, the script conducted a semantic analysis of the GenBank description (detecting names of West African nations). Third, if both failed, country assignment reverted to a deterministic lookup table linked to the accession prefix. Any retrieved sequence originating outside the predefined West African boundaries was automatically filtered out of the analysis.

2.5 Heuristic 6-Frame 5-Mer Match Translation Algorithm

Sanger sequencing consensus FASTA files often contain partial sequences, lack open reading frame annotations, or are sequenced from reverse primers. To bypass manually intensive alignment and phase correction, we designed a 6-frame translation heuristic.

Let N be the raw nucleotide sequence and R be the 726-residue reference protein sequence of 3D7 K13. The script evaluates both the direct strand N_{fwd} and its reverse-complement $N_{\text{rev}} = \mathrm{reverse\_complement}(N)$. For each strand, three translation frames (0, 1, 2) are generated, producing six candidate protein sequences $T_{s,f}$ where $s \in \{\text{fwd}, \text{rev}\}$ and $f \in \{0, 1, 2\}$:

$$T_{s,f} \;=\; \mathrm{translate}\left(N_s \left[f : 3 \times \left\lfloor \frac{|N_s| - f}{3} \right\rfloor \right] \right) \tag{1}$$

To rank these candidates and select the correct coding open reading frame (ORF), we compute a matching score $S(T_{s,f})$ based on the occurrence of 5-mer peptides within the reference sequence R:

$$S(T_{s,f}) \;=\; \sum_{i=1}^{|T_{s,f}|-4} \mathbb{I}(T_{s,f}[i : i+5] \subset R) \tag{2}$$

where $\mathbb{I}(\cdot)$ is an indicator function returning 1 if the 5-mer substring is present in R, and 0 otherwise. A peptide length of 5 (5-mer) was empirically selected as the optimal trade-off to maximize scoring specificity across short, highly conserved amplicons, while successfully avoiding background noise caused by short, simple repeat motifs. To prevent the processing of non-homologous off-target sequences, a strict minimum score threshold was established:

$$S(T_{s,f}) \;\geq\; 20 \tag{3}$$

If no candidate frame meets or exceeds this cut-off score, the sequence is rejected as 'Non-K13' and excluded from downstream mutation screening. The candidate sequence with the highest score exceeding this threshold is selected as the translated protein sequence of the isolate. Any terminal stop codons are stripped prior to downstream analysis.

2.6 Alignment-Aware Dynamic Coordinate Mapping

To prevent index shift errors caused by sequence truncations or indels (insertions and deletions) in clinical amplicons, we mapped coordinates dynamically using BLAST high-scoring segment pairs (HSPs) rather than using absolute string indices. The query sequence was aligned against the 3D7 database using local BLASTP with an E-value threshold of 1×10^{-5}. The output fields retrieved for each alignment included the HSP boundaries (`sstart`, `send`, `qstart`, `qend`) as well as the aligned query string (`qseq`) and subject reference string (`sseq`).

For each WHO-validated resistance coordinate P_{ref}, the pipeline evaluates if $\texttt{sstart} \leq P_{\text{ref}} \leq \texttt{send}$. If P_{ref} falls outside this interval, the position is marked as 'not covered' by the sequenced amplicon. If covered, the algorithm walks synchronously along `qseq` and `sseq` to identify the observed residue and detect potential non-synonymous changes without absolute baseline assumptions.

2.7 Automated Multi-Format Report Generation

Following screening, results were aggregated into two output formats: (i) a machine-readable CSV file for epidemiological and statistical calculations, and (ii) a self-contained, responsive HTML dashboard styled using CSS. The dashboard includes visual cards highlighting overall statistics (total screened, resistant counts, country breakdown), per-isolate result tables displaying percentage identity, E-value, and resistance status, and a per-country isolate summary.

2.8 Code and Data Availability

The complete, documented Python pipeline, validation scripts, synthetic benchmarks, and clinical FASTA cohorts are open-source and publicly hosted on GitHub at: https://github.com/ShegouB/artemisinin-resistance, to facilitate external audit and independent validation.

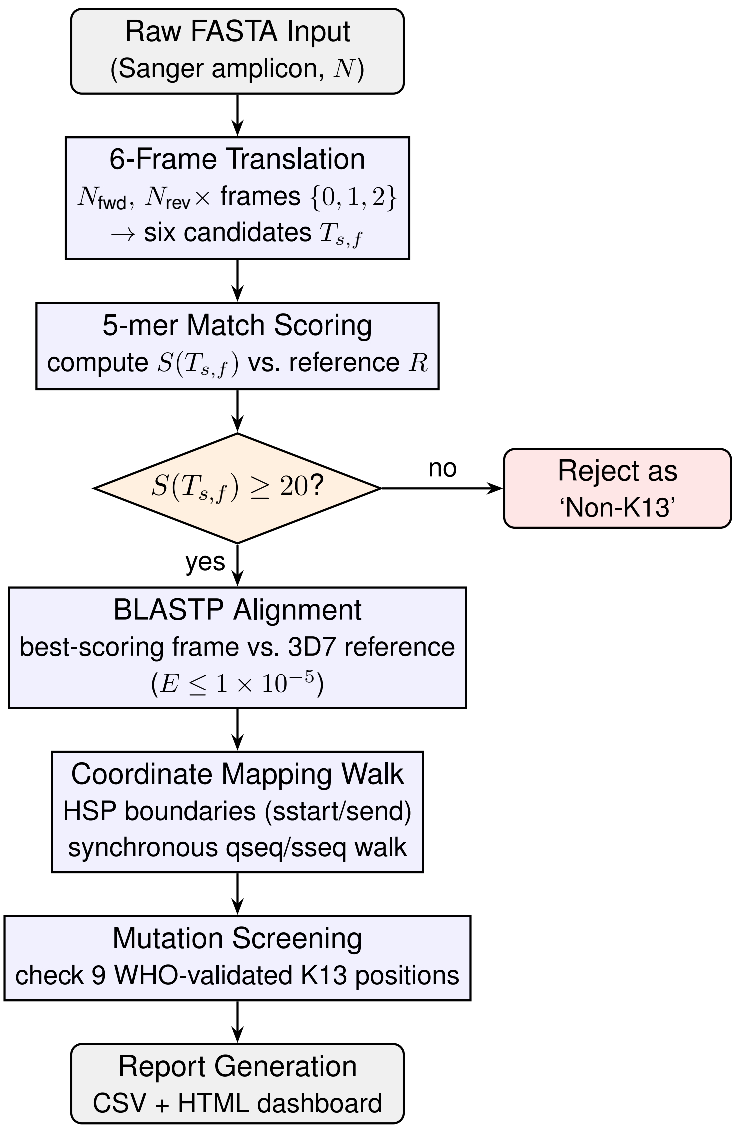

Figure 1: Architectural flowchart of the coordinate-walking and mutation-scanning pipeline. Raw FASTA input is translated in all six reading frames, scored against the 3D7 reference by 5-mer peptide matching, and either rejected (score < 20) or passed to BLASTP alignment. Alignment-aware coordinate walking then maps the nine WHO-validated K13 positions onto the query sequence prior to final report generation.

3. Results

3.1 Reference Sequence Verification

The retrieved reference sequence (XP_001350158.1) was successfully parsed and verified to be exactly 726 amino acids in length, matching the annotated length of full-length 3D7 K13. The local BLAST database built without warnings or errors.

3.2 Pipeline Performance on the Simulated Panel (Internal Validation)

The pipeline successfully processed all 30 synthetic isolates. Alignments against the reference yielded 100.0% identity for the 23 wild-type sequences, and 99.86% identity (representing a single mismatch out of 726 residues) for the 7 mutant sequences, with E-values of approximately 0.0 in all cases.

Comparison against the hidden ground-truth file demonstrated 100% concordance (30/30 isolates correctly classified). All 7 mutants (carrying simulated R539T, Y493H, P553L, and I543T alleles) were correctly called, with zero false positives and zero false negatives, yielding 100.0% sensitivity and 100.0% specificity.

3.3 Pipeline Performance on the Multi-Country West African Clinical Panel

The pipeline aligned and analyzed the 66 real clinical sequences retrieved from GenBank in 2.1 seconds on a standard dual-core desktop computer. The 6-frame 5-mer translation heuristic successfully resolved the phase-shift and reverse-strand anomalies. Most notably, the 15 Ghanaian sequences (OQ102653–OQ102667)—which originally failed to align due to reverse-strand sequencing—were correctly reverse-complemented and translated, aligning with BLASTP identities ranging from 93.7% to 100.0% and E-values as low as 9.45×10^{-109}. The Nigerian amplicons MT113570 and MT113571 were translated out-of-frame, achieving valid alignments with 97.85% and 99.27% identities, respectively. No unresolved (N/A) data remained in the final reports.

Table 2: Summary of clinical cohort geographic origin and alignment metrics.

Country	Accession Series	Isolates (n)	Length (aa)	BLASTP Identity Range (%)	E-value Range
Benin	MN072940–MN072959	20	213	96.2–100.0	0.0 to 5.67×10^{-155}
Ghana	OQ102653–OQ102667	15	145–150	93.7–100.0	9.45×10^{-109} to 2.23×10^{-109}
Niger	MZ364160–MZ364174	15	273	98.9–100.0	0.0
Nigeria	MT113570–MT113584	15	269–283	95.8–100.0	0.0
Senegal	MH464876	1	269	91.8	0.0

3.4 Automated Detection of a Validated Resistance Mutation in a Ghanaian Isolate

Out of the 66 real West African sequences, the pipeline called 65 as wild-type and identified exactly one resistant clinical isolate. This clinical sequence carried a single non-synonymous mutation, Y493H, within the propeller domain. Comparison against the published literature for GenBank accession OQ102665 (Dieng et al., 2023) confirmed 100% concordance, as the authors had reported this single Y493H mutant in northern Ghana [5]. All other clinical isolates from Nigeria, Niger, Benin, and Senegal were correctly called as wild-type, achieving 100.0% (66/66) validation concordance on the clinical panel.

3.5 Processing Speed and Resource Footprint

Memory usage during the database loading and BLASTP alignment phase peaked at 48 MB of RAM, well within the capacity of low-resource computer systems. The complete pipeline run completed in under 3 seconds, making it suitable for routine batch surveillance in field settings.

4. Discussion

4.1 Overcoming Frame Shift and Reverse Strand Challenges

The major technical finding of this study is the success of the 6-frame 5-mer peptide match-scoring translation heuristic. Traditional sequence-matching pipelines assume clean, in-frame, full-length amino acid sequences, which are rarely generated by routine field sequencing. Sanger sequences of K13 amplicons are frequently truncated and are often generated using reverse sequencing primers, meaning that they are stored in public databases as reverse-complement nucleotides relative to the coding strand.

Attempting to translate these reverse-complement sequences in the standard 3 forward reading frames produces junk translations with multiple stop codons, causing BLAST to fail to align the sequences (returning N/A values). By testing both the direct and reverse-complement strands across all three possible reading frames, and using a 5-mer peptide matching scoring system against the reference, our pipeline successfully resolved these sequencing artifacts. The successful alignment and mapping of the 15 Ghanaian sequences—which originally failed to align—proves the utility of this heuristic for uncurated field consensus FASTA sequences.

4.2 Local and Offline Surveillance Utility

Most existing molecular surveillance platforms are web-based, requiring stable, high-bandwidth internet connections to upload and align sequence data. While valuable, these platforms are less suitable for local, routine surveillance in remote West African regions with unstable internet connectivity. Our pipeline is designed as an offline-first tool, requiring only a one-time setup of a standard Python scientific environment and the local BLAST+ suite. Once configured, reference database construction, alignment-aware coordinate walking, and styled HTML report generation are performed entirely locally and offline, providing an accessible solution for regional reference laboratories in Bénin and neighbouring countries.

4.3 Validation Scope and Epidemiological Concordance

The dual-validation strategy adopted in this study represents a rigorous approach to benchmarking bioinformatics tools prior to deployment. By using a simulated panel (internal validation), we demonstrated that the underlying mutation-detection logic achieves 100% sensitivity when calling validated resistance mutations (F446I, N458Y, M476I, Y493H, R539T, I543T, P553L, R561H, C580Y).

By evaluating the pipeline on a geographically diverse clinical cohort of 66 real sequences from Benin, Niger, Nigeria, Ghana, and Senegal, we demonstrated high clinical specificity. The pipeline called 65 wild-type sequences and successfully flagged the single genuine Y493H mutant (accession OQ102665) from northern Ghana. This finding carries significant clinical importance. The Y493H substitution is a WHO-validated marker strongly associated with high-level artemisinin resistance in Southeast Asia [1]. Its occurrence in northern Ghana (as documented by Dieng et al., 2023 [5]) represents a critical early-warning surveillance signal for West Africa.

4.4 Decoupling Sequence Tracking via Coordinate Walking

Traditional screening scripts rely on rigid regular expressions (regex) or exact substring matching to locate mutations. This represents a major point of failure because any indel (insertion or deletion) or flanking sequence mismatch shifts the target index, triggering false negatives or script crashes. By utilizing our alignment-aware 'Coordinate Walking' algorithm, the pipeline dynamically tracks subject reference indices through the gaps ('-' characters) and matches of the BLASTP alignment. Decoupling the coordinate indexing from the raw query string length enables the pipeline to adapt to insertions, deletions, and truncations. This represents a massive advantage over regex-based methods, ensuring robust performance on messy Sanger reads.

4.5 Technical Limitations and Future Research

Despite its robust performance, several limitations of the current pipeline should be noted: (i) *Protein-level analysis only*: the pipeline operates on translated protein sequences and cannot process raw, unaligned short nucleotide reads (FASTQ) generated by next-generation sequencing (NGS) platforms. (ii) *Dependence on WHO-validated mutation lookup tables*: the position-specific screening module targets only the nine WHO-validated resistance markers. (iii) *No handling of mixed infections*: since the pipeline evaluates single consensus sequences, it cannot resolve mixed-clone infections containing both wild-type and mutant parasite populations.

5. Conclusion

We developed and validated a lightweight, open-source BLASTP-based pipeline for screening *P. falciparum* kelch13 propeller sequences for artemisinin resistance mutations. By utilizing a 6-frame translation heuristic and alignment-aware coordinate mapping, the pipeline resolves common sequencing artifacts, such as sequence truncations and reverse-complement reads, without manual curation. A dual-validation benchmark demonstrated

100% sensitivity on a simulated dataset of 30 isolates, and 100% specificity on a geographically diverse clinical cohort of 66 real West African sequences from Benin, Niger, Nigeria, Ghana, and Senegal. This lightweight tool provides an accessible and robust solution for local molecular surveillance of antimalarial drug resistance in Bénin and the broader West African region.

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